**1** Prove algebraically that the sum of any two consecutive integers is always an odd number.

$n + n+1$

$2n + 1$

even

even + 1 is odd

**(Total for question 1 is 2 marks)**

**2** Prove algebraically that the sum of any three consecutive even integers is always a multiple of 6.

$2n + 2n+2 + 2n + 4$

$6n + 6$

$6(n+1)$

**(Total for question 2 is 2 marks)**

**3** Prove that $(3n + 1)^2 - (3n - 1)^2$ is always a multiple of 12, for all positive integer values of $n$.

$$((3n+1)(3n+1)) - ((3n-1)(3n-1))$$

$$(9n^2 + 3n + 3n + 1) - (9n^2 - 3n - 3n + 1)$$

$$(9n^2 + 6n + 1) - (9n^2 - 6n + 1)$$

$$9n^2 + 6n + 1 - 9n^2 + 6n - 1$$

$$\underline{\underline{12n}}$$

**(Total for question 3 is 2 marks)**

**4** $n$ is an integer.
Prove algebraically that the sum of $n(n + 1)$ and $n + 1$ is always a square number.

$$n(n+1) + n+1$$

$$n^2 + n + n + 1$$

$$n^2 + 2n + 1$$

$$(n+1)(n+1)$$

$$\underline{\underline{(n+1)^2}}$$

**(Total for question 4 is 2 marks)**

**5** Prove that $(2n + 3)^2 - (2n - 3)^2$ is always a multiple of 12, for all positive integer values of $n$.

$((2n+3)(2n+3)) - ((2n-3)(2n-3))$

$(4n^2+6n+6n+9) - (4n^2-6n-6n+9)$

$(4n^2+12n+9) - (4n^2-12n+9)$

$4n^2+12n+9 - 4n^2+12n-9$

$24n$

$12(2n)$

**(Total for question 5 is 2 marks)**

**6** $n$ is an integer.
Prove algebraically that the sum of $(n + 2)(n + 1)$ and $n + 2$ is always a square number.

$(n+2)(n+1) + n+2$

$n^2+n+2n+2+n+2$

$n^2+4n+4$

$(n+2)(n+2)$

$(n+2)^2$

**(Total for question 6 is 2 marks)**

**7** Prove that the sum of 3 consecutive odd numbers is always a multiple of 3.

$2n + 1 + 2n + 3 + 2n + 5$

$6n + 9$

$3(2n + 3)$

**(Total for question 7 is 2 marks)**

**8** Prove that the sum of 3 consecutive even numbers is always a multiple of 6.

$2n + 2n + 2 + 2n + 4$

$6n + 6$

$\cancel{2(3n + 3)}$

$6(n + 1)$

**(Total for question 8 is 2 marks)**

**9** Prove algebraically that the sum of the squares of any 2 even positive integers is always a multiple of 4.

$$(2n)^2 + (2m)^2$$

$$4n^2 + 4m^2$$

$$4(n^2 + m^2)$$

**(Total for question 9 is 2 marks)**

**10** Prove algebraically that the sum of the squares of any 2 odd positive integers is always even.

$$(2n+1)^2 + (2m+1)^2$$

$$(2n+1)(2n+1) + (2m+1)(2m+1)$$

$$4n^2 + 2n + 2n + 1 + 4m^2 + 2m + 2m + 1$$

$$4n^2 + 4n + 4m^2 + 4m + 2$$

$$2(2n^2 + 2n + 2m^2 + 2m + 1)$$

**(Total for question 10 is 2 marks)**

**11** Prove that the sum of the squares of any two consecutive integers is always an odd number.

$$n^2 + (n+1)^2$$

$$n^2 + (n+1)(n+1)$$

$$n^2 + n^2 + n + n + 1$$

$$n^2 + n^2 + 2n + 1$$

$$2n^2 + 2n + 1$$

$$\underbrace{2(n^2 + n)}_{\text{even}} + 1$$

even + 1 is odd.

**(Total for question 11 is 3 marks)**

**12** Prove that the sum of the squares of two consecutive odd numbers is always 2 more than a multiple of 8

$$(2n+1)^2 + (2n+3)^2$$

$$(2n+1)(2n+1) + (2n+3)(2n+3)$$

$$4n^2 + 2n + 2n + 1 + 4n^2 + 6n + 6n + 9$$

$$8n^2 + 16n + 10$$

$$8n^2 + 16n + 8 + 2$$

$$\underbrace{8(n^2 + 2n + 1)}_{\text{multiple of } 8} \underbrace{+ 2}_{+2}$$

**(Total for question 12 is 2 marks)**

**13** Prove that the difference between the squares of any 2 consecutive integers is equal to the sum of these integers.

$n$ and $n+1$

$n+n+1 = 2n+1$

$(n+1)^2 - n^2$

$(n+1)(n+1) - n^2$

$n^2+n+n+1 - n^2$

$2n+1$

**(Total for question 13 is 3 marks)**

**14** Prove algebraically that the sums of the squares of any 2 consecutive even number is always 4 more than a multiple of 8.

$(2n)^2 + (2n+2)^2$

$4n^2 + (2n+2)(2n+2)$

$4n^2 + 4n^2 + 4n + 4n + 4$

$\cancel{4}\ 8n^2 + 8n + 4$

$\underbrace{8(n^2+n)}_{\text{multiple of 8}} \underbrace{+4}_{+4}$

**(Total for question 14 is 3 marks)**